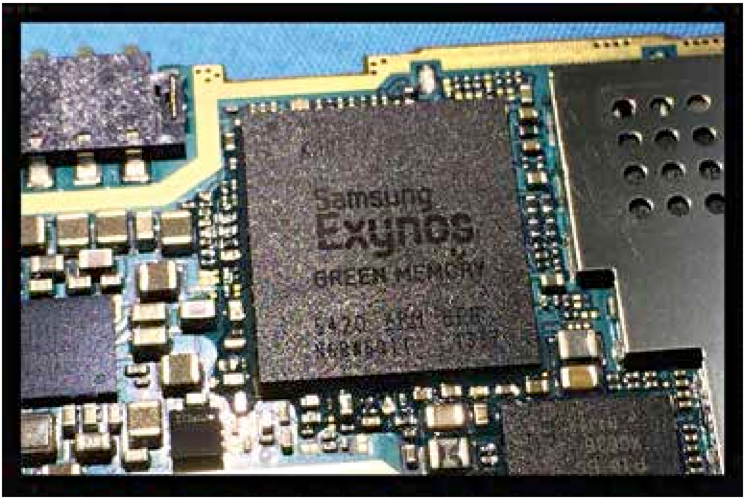
processes down to 20nm. However, after this was achieved, the integration started to crumble.

Due to this scaling down, when the voltage needed for one process was delivered to the chip, the other processes stopped working properly. To solve this issue, FinFET technology was developed.

Way Forward For SoCs

Intel, Qualcomm, and Samsung have all announced their own 10nm FinFET chips, which would reduce the process size even further from the current 14nm System-on-Chips.



An Exynos Chipset Showing the transistors and arrangement of processes.

According to Samsung, which has developed the Exynos 8895 SoC (a 10nm chip), the decrease in process size has allowed it to house 30 percent more transistors in the same area than previous gen 14nm chips.

With the manufacture of 10nm chips, developers will be able to improve performance by 27 percent, or decrease power consumption by 40 percent in devices.

However, one has to wonder how much smaller can process size get before affecting the overall functionality of the SoC. There have been reports that already with 10nm, developers are facing several issues regarding the architecture.